1) The Mean and standard deviation of the form marks obtained by 1000 students in examination are respectively 34.5 & 16.5 assuming the normality of distribution find the approximate no. of students Expected to obtain marks 30 and 60.

Sol^{n} Given that $\mu = 34.5$ and $\sigma = 16.5$

then $P(30 \leq x \leq 60) = ?$

Now, when $x = 30 \Rightarrow z = \frac{x-\mu}{\sigma} = \frac{30-34.5}{16.5} = -0.27$

$= z_1$ (say)

when $x = 60 \Rightarrow z = \frac{x-\mu}{\sigma} = \frac{60-34.5}{16.5} = 1.54 = z_2$ (say)

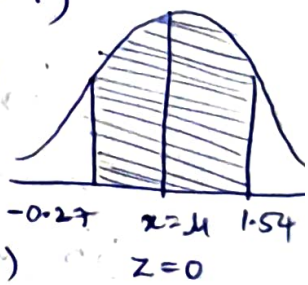
$\therefore P(30 \leq x \leq 60) = P(-0.27 \leq z \leq 1.54)$

$= P(-0.27 \leq z \leq 0) + P(0 \leq z \leq 1.54)$

$= P(0 \leq z \leq 0.27) + P(0 \leq z \leq 1.54)$

$= 0.1064 + 0.4382$ ($\because$ By symmetry)

$= 0.5916$



$\therefore$ No. of student who get marks b/n 30-60

$= 0.5916 \times 1000.$

$= 546.6$

$\simeq 547$

$\therefore$ Hence, 547 students get marks b/n 30-60.

2) Marks obtained in mathematics by 1000 students is normally distributed with mean 78% and SD 11%. Determine.

a) How many students get marks above 90%.

b) ~~mark~~. what was the highest mark obtained by the lowest 10% of students

c) within what limits did the middle of 90% of the students lie.

Soln $\mu = 78\% = 0.78$; $\sigma = 11\% = 0.11$

(i) $p(X > 90\%) = p(x > 0.9)$

when $x = 0.9 \Rightarrow z = \frac{x-\mu}{\sigma} = \frac{0.9-0.78}{0.11} = 1.09$.

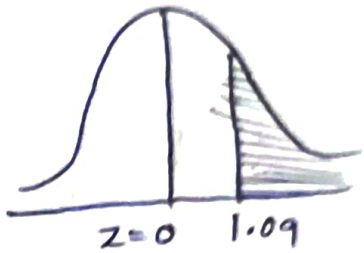
$\therefore p(x > 90\%) = p(x > 0.9)$

$= p(z > 1.09)$

$= 0.5 - p(0 < z < 1.09)$

$= 0.5 - 0.3621$

$= 0.1379$

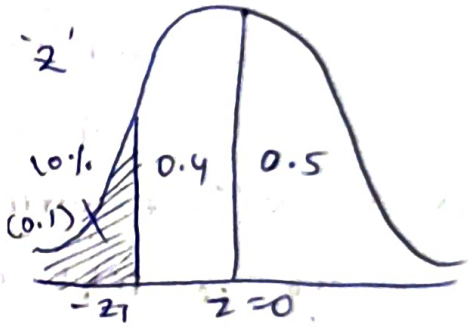


$\therefore$ Hence the no. of students with marks more than 90%

$= 0.1379 \times 1000$

$= 137.9\% \approx 138\%$

b) The 0.1 area to the left of 'z' corresponds to the lowest 10% of the students.



From the figure, we have

probability of $p(-z_1 < z < 0) = 0.4$

$p(0 < z < z_1) = 0.4$

$z_1 = 1.28$

$\therefore z_1 = -1.28$ (left side).

Now; $z = \frac{x-\mu}{\sigma} = z_1$

$\Rightarrow \frac{x-0.78}{0.11} = -1.28$

$x = -1.28 \times 0.11 + 0.78$

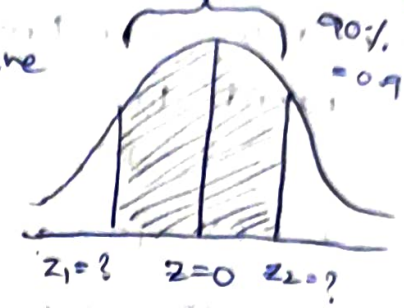
$= 0.6392$ (or) $x = 63.92\%$

-highest mark obtained by lowest 10% of the student $= 0.63$

$= 63.92\%$

$\simeq 64\%$.

c) From the above figure, we have



$$P(0<z<z_2) = 0.45$$

$$\Rightarrow z_2 = 1.64$$

and $P(-z_1 < z < 0) = 0.45$

$$\Rightarrow z_1 = -1.64$$

Now, $z = \frac{x-\mu}{\sigma} = z_1$

$$\frac{x-0.78}{0.11} = -1.64$$

$$\Rightarrow x_1 = -1.64 \times 0.11 + 0.78$$

$$x_1 = 0.5996 = 59.96\% = 60\%.$$

and $z = \frac{x-\mu}{\sigma} = z_2$

$$\Rightarrow \frac{x-0.78}{0.11} = 1.64$$

$$x = 1.64 \times 0.11 + 0.78 = 0.9604$$

$$x = 96.04\% \Rightarrow 96\%.$$

3) If 'x' is a normal variance find the area A

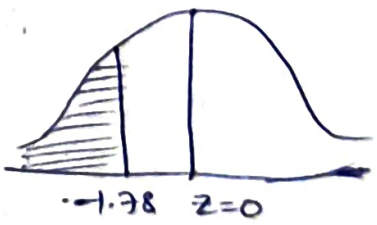
i) To the left of $z = -1.78$

ii) To the Right of $z = -1.45$

iii) Correspondias to $-0.8 \leq z \leq 1.53$

iv) To the left of $z = -2.52$ and to the right of $z = 1.83$.

Sol'r i) The required area 'A'



$$= 0.5 - P(-1.78 < z < 0)$$

$$= 0.5 - P(0 < z < 1.78)$$

$$= 0.5 - 0.4625$$

$$= 0.0375$$

ii) The required area 'A'

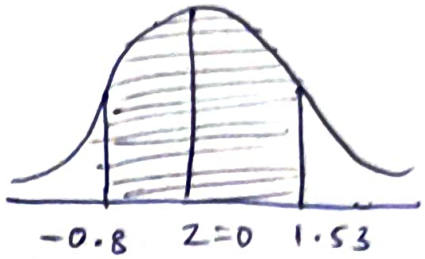
$$= 0.5 + P(-1.45 < z < 0)$$

$$= 0.5 + P(0 < z < 1.45)$$

$$= 0.5 + 0.4265$$

$= 0.9265$

iii) $P(-0.8 \leq z \leq 1.53)$

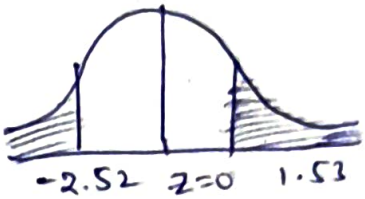


$= P(-0.8 \leq z \leq 0) + P(0 \leq z \leq 1.53)$

$= 0.2881 + 0.4370$

$= 0.9251$

iv) The required are 'A'



$= [0.5 - P(-2.52 < z < 0)] +$

$[0.5 - P(0 < z < 1.83)]$

$= [0.5 - P(0 < z < 2.52)] + [0.5 - 0.4664]$

$= [0.5 - 0.4941] + [0.5 - 0.4664]$

$= 0.0395$

• Sampling Distribution

population: population is the set or collection of objects under the study

size: No. of population in the observation.

denoted by 'N'. (or)

- No. of objects or obeservation in the population is the size of the population

Sample: The finite subset of the population is know as Sample (or)

A small section or part selected from the population is called a size of the sample.

It is denoted by 'n'.

population	Sample
1) Eng. students in Telangana	1) Eng. students in UCEJ
2) Total products of items in a month	2) Total produ^{n} of items in a day.

Sampling: The process of drawing a sample.

Large sampling: The sampling is said to be large sampling if $n \geq 30$.

Small sampling: The sampling is said to be small sampling if $n \leq 30$.

Sampling with replacement:

It means each member of the population may be choosen morethan once.

∴ Total no. of Samples with replacement is N^{n}

where N = no. of population size

n = no. of sample size.

Sampling without Replacement :-

It means each no. of population can be choosen only once

∴ Total no. of Samples without Replacement is $^{N}C_{n}$

Random Sampling :-

It is a sampling in which each member of the population has equal chances or equal propability is being included in the sampling.

- The sample obtained by this method is known as Random sampling.

Parameters :-

statical constants obtained from the population are known as population parameters (or) population parameters on parameters

Ex:- mean = μ

Variance = σ^2

Standard deviation = σ

Statics :- Statical constants obtained from the Sample observations are known as Statastics.

Ex:- Sample mean = $\mu_{\bar{x}}$

Sample variance = s^2

Sample SD = s

Sample propotion = p

★★★ :- A population consists of five members 2, 3, 6, 8 and 11. consider all possible samples of size two. which can be drawon with replacement from this population. find

a) The mean of the population.

b) The SD of the population

c) The mean of the sampling distribution of means

d) The SD of the sampling distribution of mean.

a) Given population is 2, 3, 6, 8 and 11

Mean of the population is given by

$$\mu = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6.$$

b) Variance of the population is given by

$$\sigma^2 = \frac{1}{n}\{(x_i - \bar{x})^2$$

$\bar{x}$ = mean.

n = sample size

$$= \frac{1}{5}(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2$$

$$= \frac{1}{5}(-4)^2 + (-3)^2 + 0 + (2)^2 + (5)^2$$

$$= 10.8$$

c) The SD of the population is $\sqrt{10.8} = 3.29$

$$\sigma = 3.29$$

c) Sampling with replacement (Infinite pop.)

The total no. of samples with replacement is

$$N^n = 5^2 = 25$$

N = pop. size

n = sample size

listing all possible samples from size 2 from the population 2, 3, 6, 8, 11 with replacement

$$\left\{\begin{array}{l} (2,2), (2,3), (2,6), (2,8), (2,11) \\ (3,2), (3,3), (3,6), (3,8), (3,11) \\ (6,2), (6,3), (6,6), (6,8), (6,11) \\ (8,2), (8,3), (8,6), (8,8), (8,11) \\ (11,2), (11,3), (11,6), (11,8), (11,11) \end{array}\right\} \downarrow \text{Samples}$$

Their corresponding means are

$\frac{2+2}{2} = \frac{4}{2} = 2$

$\frac{2+3}{2} = \frac{5}{2} = 2.5$

$\frac{2+9}{2} = \frac{8}{2} = 4$

$$\left\{\begin{array}{ccccc} 2 & 2.5 & 4 & 5 & 6.5 \\ 2.5 & 3 & 4.5 & 5.5 & 7 \\ 4 & 4.5 & 6 & 7 & 8.5 \\ 5 & 5.5 & 7 & 8 & 9.5 \\ 6.5 & 7 & 8.5 & 9.5 & 11 \end{array}\right\} \text{①}$$

Sampling distribution of means.

∴ The mean of Sampling Distribution of means

$$M_{\bar{x}} = \frac{2+2.5+4+5+\cdots\cdots 11}{25} = \frac{150}{25} = 6.$$

$$M_{\bar{x}} = \bar{X} = 6.$$

d) The variance of the sampling distribution of means

$$S^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$$

$$= \frac{1}{25}\left[(2-6)^2 + (2.5-6)^2 + (4-6)^2 + \cdots\cdots + (11-6)^2\right]$$

$$= \frac{1}{25}\left[\frac{135}{25}\right] = 5.40$$

$$S^2 = 5.40 \Rightarrow S = \sqrt{5.40}$$

$$S = 2.32$$

Sampling Distribution :-

Consider a population of size 'N' and draw all possible samples of size 'n' at random for each sample we can conclude the mean.

and the mean of the sample will not be Identical

→ If the group this different means according to their frequencies, the frequency dist^n is formed is known as sampling distribution of the mean.

Similarly, we can have Sampling distribution of the SD etc.

→ S.D is also called standard Error.

Infinite population :-

Infinite population with replacement – σ Known

let $x_1, x_2, \ldots x_n$ be the 'n' Random samples drawn from a population of the size 'N', with mean μ & variance σ^2 and $\bar{x}$ is the mean of the samples.

* (i) The mean of the sampling distribution of means :-

we have $\bar{x} = \dfrac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$

$$E(\bar{x}) = E\left[\frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}\right]$$

$$\mu_{\bar{x}} = \frac{1}{n}\left[E(x_1) + E(x_2) + E(x_3) + \ldots + E(x_n)\right]$$

$$E(\bar{x}) = \mu_{\bar{x}} = \frac{1}{n}\left[\mu + \mu + \mu + \ldots + \mu\right]$$

$$= \frac{1}{n}\left[n\mu\right] = \mu$$

$$\boxed{\mu_{\bar{x}} = \mu}$$

* ii) Variance

we have $\bar{x} = \dfrac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$

$$V(\bar{x}) = V\left(\frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}\right)$$

$$= \frac{1}{n^2}\left[V(x_1) + V(x_2) + V(x_3) + \ldots + V(x_n)\right]$$

$\because$ let $V(x_i) = \sigma^2$ (say)

$$V(\bar{x}) = \frac{1}{n^2}\left[\sigma^2 + \sigma^2 + \sigma^2 \ldots \sigma^2\right]$$

$$= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$V(\bar{x}) = \frac{\sigma^2}{n} \text{ (or) } \boxed{\sigma^2_{\bar{x}} = \frac{\sigma^2}{n}}$$

and $\boxed{\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}}$

$$\mu_{\bar{x}} = \mu$$
$$\sigma_{\bar{x}} = S = \frac{\sigma}{\sqrt{n}}$$
} sampling is done with replacement

Infi. Finite population without replacement - σ known

$$\mu_{\bar{x}} = \mu$$
$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$
} Sampling is done without replacement

Here $\frac{N-n}{N-1}$ is known as finite population correction factor.

Standard Error:

The SD of the sampling distribution of means (SE)

Formulae of SE:

a) SE of sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$

b) S.E of same proportion $(p) = \sqrt{\frac{PQ}{N}}$ where $Q = 1-P$

c) SE of sample SD $(S) = \frac{\sigma}{\sqrt{2n}}$

d) SE of difference of two sample means $\bar{x}_1$ and $\bar{x}_2$

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

e) SE of difference of two sample propotions

$$SE(P_1 - P_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

f) SE of sample SD : $SE(S_1 - S_2) = \sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$

① Find the value of finite population correction factor for $n = 10$ and $N = 1000$.

Sol:

N = Size of the population = 1000

n = Size of sample = 10

∴ population correction factor $= \frac{N-n}{N-1}$

$= \frac{1000-10}{1000-1} = \frac{990}{999}$

② A population consists of 5, 10, 14, 18, 13 and 24 consider all possible samples of size '2' which can be drawn without replacement from the population. and find

a) Mean of the population

b) The SD of the population

c) The mean of the sample dist^n of means.

d) The SD of " " " mean.

③ 1, 2, 3, 4, 5, 6 with replacement a, b, c, d.

H/W ↓ and without replacement. (sample size is 2)

Questions.

③ A population consists of 1, 2, 3, 4, 5 and 6 consider all possible samples of size '2' which can be drawn without replacement from population & with replacement. a) Mean of the population

b) The SD of the populn.

c) The mean of sample dist^n of means

d) The SD of " " of mean

3 Solⁿ a) Given population is 1, 2, 3, 4, 5, 6

Mean of the population is given by

$$\mu = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

b) variance of the population is given by

$$\sigma^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$$

$$\sigma^2 = (1-3.5)^2 + (2-3.5)^2 + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2 + (6-3.5)^2$$

$$\sigma^2 = \frac{(-2.5)^2 + (-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2 + (2.5)^2}{6}$$

$$\sigma^2 = \frac{6.25 + 2.25 + 0.25 + 2.25 + 6.25}{6} = \frac{17.5}{6} \simeq 2.917$$

$$SD = \sigma = \sqrt{2.917} = 1.708$$

c) The mean of the sample distribution of means

The total no. of samples with replacement is

$$N^n = 6^2 = 36$$

listing all possible samples from size '2' from the population 1, 2, 3, 4, 5, 6 with replacement

$$\begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

Their corresponding means are

$$\begin{Bmatrix} 1 & 1.5 & 2 & 2.5 & 3 & 3.5 \\ 1.5 & 2 & 2.5 & 3 & 3.5 & 4 \\ 2 & 2.5 & 3 & 3.5 & 4 & 4.5 \\ 2.5 & 3 & 3.5 & 4 & 4.5 & 5 \\ 3 & 3.5 & 4 & 4.5 & 5 & 5.5 \\ 3.5 & 4 & 4.5 & 5.0 & 5.5 & 6 \end{Bmatrix}$$

The mean of the distribution is

$$\mu_{\bar{x}} = \frac{1 + 1.5 + 2 + 2.5 \text{ ------- } + 6}{36}$$

$$\mu_{\bar{x}} = 3.5$$

d) The variance of the sampling distribution of mean.

$$s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{36} \left[(1-3.5)^2 + (2-3.5)^2 + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2 + (6-3.5)^2 \right]$$

2

$$\frac{\sigma^2}{n} = \frac{9.9167}{2} = 1.4583.$$

$$\sigma^2 = 1.4583 \Rightarrow \sigma = \sqrt{1.4583}$$

$$\sigma = 1.2076,$$

Without replacement :

a) Mean of the population

$$\mu = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

b) S.D of population

$$\sigma^2 = \frac{17.5}{6} \simeq 2.917$$

$$\sigma = \sqrt{2.917} = 1.708$$

Similar to with replacement (mean & variance)

c) Sampling distribution of sample means.

$n = 2$; $N = 6$

$$NC_n = {}^6C_2 = 15$$

$$\mu_{\bar{x}} = \frac{\sum \bar{x}}{NC_n} = \frac{52.5}{15} \rightarrow 1.5 + 2 + 2.5 + 3 + 3.5 + \cdots - 6$$

$$\mu_{\bar{x}} = 3.5$$

d) SD of sampling distribution.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$\sigma_{\bar{x}} = \frac{1.708}{\sqrt{2}} \sqrt{\frac{6-2}{6-1}}$$

$$= 1.708 \times \frac{1}{\sqrt{2}} \times \sqrt{\frac{4}{5}}$$

$$\boxed{\sigma_{\bar{x}} \simeq 1.080}$$

4) A population consists of 2, 3, 6, 8, 11 — consider all possible sample of size is 2, which can be drawn without replacement from the population.

a) Mean of population.

b) S D of population

c) the mean of sample distribution of mean

d) S.D of sample distribution of mean.

Sol: Given population is 2, 3, 6, 8, 11

a) mean of population is given by

$$\mu = \frac{2+3+6+8+11}{5} = 6$$

$$\boxed{\mu = 6}$$

b) S.D of the population

$$\sigma = \sqrt{\frac{1}{n}\sum (x_i - \mu)^2}$$

$$= \sqrt{\frac{1}{5}[2-6]^2 + [3-6]^2 + [8-6]^2 + [11-6]^2}$$

$$\sigma = 3.28$$

c) no. of sample of size '2' without replacement is

$$^{N}C_n = {}^{5}C_2 = 10$$

listing all possible sampling of size from the given population without replacement.

$$\left\{ \begin{array}{l} (2,3), (2,6), (2,8), (2,11) \\ (3,6), (3,8), (3,11) \\ (6,8), (6,11) \\ (8,11) \end{array} \right\}$$

Their corresponding mean.

$$\left\{ \begin{array}{l} 2.5, \quad 4, \quad 5, \quad 6.5 \\ 4.5 \quad 5.5 \quad 7 \\ 7 \quad 8.5 \\ 9.5 \end{array} \right\}$$

the mean of the distribution of mean

$$\mu = \mu_{\bar{x}} = \frac{2.5 + 4 + 5 + 6.5 + 4.5 + \cdots + 9.5}{10}$$

$$\mu_{\bar{x}} = \frac{60}{10}$$

$$\mu = \mu_{\bar{x}} = 6.$$

d) Variance of the sampling distribution of means

$$\sigma_{\bar{x}}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$\sigma_{\bar{x}}^2 = \frac{1}{10} \left[(2.5-6)^2 + (4-6)^2 + (5-6)^2 + \cdots (8.5-6)^2 + (9.5-6)^2 \right]$$

$$\sigma_{\bar{x}}^2 = 4.05 \Rightarrow \sigma_{\bar{x}} = \sqrt{4.05}$$

$$\sigma_{\bar{x}} = 2.01$$

Verification:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{3.28}{\sqrt{2}} \sqrt{\frac{5-2}{5-1}}$$

$$\sigma_{\bar{x}} = 2.008$$

5) Variance of population is 2. The size of sample collected from the population is 169. what is the standard error of means.

Soln- Standard error of means $(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

Given that,

Variance $(\sigma^2) = 2$

$$\sigma = \sqrt{2}$$

Sample size $(n) = 169$

S.E of $(\bar{x})$ is $\frac{\sqrt{2}}{\sqrt{169}} = \frac{\sqrt{2}}{13}$

$$\bar{x} = 0.10846.$$

6) what is effect on S.E of the sample take from an infinite population of sample size increasing from 400 to 900.

Solr Given that $n_1 = 400$; $n_2 = 900$

$$S.E = \frac{\sigma}{\sqrt{n_1}} = \frac{\sigma}{\sqrt{400}} = \frac{\sigma}{20} \quad \text{—①}$$

$$S.E = \frac{\sigma}{\sqrt{n_2}} = \frac{\sigma}{\sqrt{900}} = \frac{\sigma}{30} \quad \text{—②}$$

Now; $SE_1 = \frac{\sigma}{20} \Rightarrow \frac{\sigma}{2 \times 10}$

$$= \frac{3\sigma}{2 \times 30}$$

$$SE_1 = \frac{3}{2} SE_2$$

7) when a sample take from any finite population what happened to the S.E after mean, if sample size decreare from 800 to 200.

Solr Given that $n_1 = 800$; $n_2 = 200$.

$$SE = \frac{\sigma}{\sqrt{n_1}} = \frac{\sigma}{\sqrt{800}} = \frac{\sigma}{20\sqrt{2}} \quad \text{—①}$$

$$SE = \frac{\sigma}{\sqrt{n_2}} = \frac{\sigma}{\sqrt{200}} = \frac{\sigma}{10\sqrt{2}} \quad \text{—②}$$

Now; $SE_1 = \frac{\sigma}{20\sqrt{2}} \Rightarrow \frac{\sigma}{2 \times 10\sqrt{2}}$

$$SE_1 \Rightarrow \frac{1}{2} SE_2$$

8) A random sample of 100 packages skipped by all flighting 13 has some damages. Find the S.E of proportion. proportion of damage pack

$$P = \left(\frac{13}{100}\right)$$

Soln proportion of damage package (p) $= \frac{13}{100}$

$= 0.13$

$Q = 1 - p$

$Q = 1 - 0.13 = 0.87$

$\boxed{Q = 0.87}$

and $n = 100$

$SE(P) = \sqrt{\frac{PQ}{n}}$

$= \sqrt{\frac{(0.13)(0.87)}{100}}$

$SE(P) = 0.0337$

** ★ 9) Let $\mu_1 = (3, 7, 8)$; $\mu_2 = (2, 4)$ find (a)

(a) μ_{μ_1}

(b) μ_{μ_2}

(c) Mean of sampling distribution of the differen of means $\mu_{\mu_1} - \mu_2$

(d) σ_{μ_1}

(e) σ_{μ_2}

(f) $\sigma_{\mu_1 - \mu_2}$

Soln Given that $\mu = (3, 7, 8)$, $\mu_2 = (2, 4)$

$\mu_1 - \mu_2 = \{3-2, 7-2, 8-2, 3-4, 7-4, 8-4\}$

$= \{1, 5, 6, -1, 3, 4\}$

$= \{-1, 1, 3, 4, 5, 6\}$ Ascending

a) $\mu_{\mu_1} = \frac{3+7+8}{3} = \frac{18}{3} = 6$

(b) $\mu_{\mu_2} = \frac{2+4}{2} = \frac{6}{2} = 3$

(c) $\mu_{\mu_1 - \mu_2} = \frac{-1+1+3+4+5+6}{6} = \frac{18}{6} = 3$

d) $\sigma_{\mu_1} = \sqrt{\frac{1}{n}\sum(x_i - \mu_{\mu_1})^2}$

$= \sqrt{\frac{1}{3}(3-6)^2 + (7-6)^2 + (8-6)^2}$

$\sigma_{\mu_1} = \sqrt{\frac{14}{3}}$

$\sigma_{\mu_1} = 2.1602$

e) $\sigma_{\mu_2} = \sqrt{\frac{1}{2}[(2-6)^2 + (4-6)^2]}$

$= \sqrt{\frac{1}{2}[(1-4)^2 + (-2)^2}$

$= \sqrt{\frac{20}{2}} \; 10 = \sqrt{10}$

f) $\sigma_{\mu_1 - \mu_2} = \sqrt{\frac{1}{n}\sum(x_1 - \mu_{\mu_1} - \mu_2)^2}$

$= \sqrt{\frac{1}{5}[(3-6)^2 + (7-6)^2 + (8-6)^2 - (2-6)^2 - (4-6)^2}$

$= \sqrt{\frac{1}{5}(-3)^2 + (1)^2 + (4) - (-4)^2 - (-2)^2}$

$\sqrt{\frac{1}{5}}\sqrt{9+1+4+16+4}$

$\sqrt{\frac{1}{5}}\sqrt{34} \Rightarrow \sqrt{\frac{34}{5}}$

Central Limit Theorem

If $\bar{x}$ be of a sample size 'n' of a population with μ and SD σ. then the standardized sample mean.

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$ is a random variable.

whose distribution function approaches that of the standard normal distribution.

$$N(z, 0, 1) \text{ as } n \to \infty$$

Q) The mean high of a students in a college is 155 cm the SD is 15 what is the probability that the mean of 36 student is lessthan 157 cm.

Sol:- Mean = 155 cm = μ (mean of the height of the size).

$\sigma = 15$ = SD of the population.

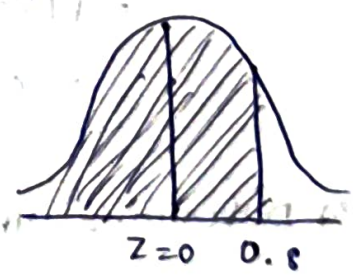
$n = 36$

$\bar{x} = 157$ cm

$P(\bar{x} < 157) = ?$

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{157 - 155}{\frac{15}{\sqrt{36}}} = 0.8$$



$P(\bar{x} \leq 157) = P(z < 0.8)$

$= 0.5 + P(0 < z < 0.8)$

$= 0.5 + 0.2881 = 0.7881$

Thus the probability that the mean height of 36 student is lessthan 157.

$\Rightarrow 0.7881$

2) A Random sample of size 100 which taken from an infinite population having the mean $\mu = 76$ and variance $\sigma^2 = 256$. what is the probability that $\bar{x}$ will be b/n 75 and 78.

Soln $P(75 \le \bar{x} \le 78) = ?$

Given $\mu = 76 \rightarrow$ mean of population

$\sigma^2 = 256$

$\sigma = 16$

$n = 10$. (size)

let $z = \dfrac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$\bar{x} = 75 \Rightarrow z = \dfrac{75 - 76}{\frac{16}{10}} = \dfrac{-10}{1.6} = -0.625$

$\bar{x} = 78 \rightarrow z = \dfrac{78 - 76}{\frac{16}{10}} = \dfrac{20}{16} = \dfrac{5}{4} = 1.25$

$P(-0.625 \le z \le 1.25)$

$P(-0.625 \le z \le 0) + P(0 \le z \le 1.25)$

$= P(0 \le z \le 0.625) + P(0 \le z \le 1.25)$

$= 0.2324 + 0.3944$

$= 0.6268.$

3) Determine the expected no. of random samples having their means

a) B/n 22.39 and 22.41

b) greaterthan 22.42

c) lessthan 22.37

d) lessthan 22.38 or morethan 22.41

for the following data.

Sol'r N = Size of the population = 1500.

n = Size of the sample = 36.

σ = population S.D = 0.48

μ = population mean = 22.40

(a) $P(22.39 < \bar{x} < 22.41) = ?$

let $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \Rightarrow z = 22.39$

$$\Rightarrow \frac{22.39 - 22.40}{\frac{0.48}{\sqrt{36}}}$$

$$= \frac{22.39 - 22.40}{\frac{0.48}{6}} = \frac{-0.01}{0.08} = -0.125$$

Now, $\bar{x} = 22.41$

$$\Rightarrow \frac{22.41 - 22.40}{0.08} = \frac{0.01}{0.08} = 0.125$$





UNIT-IV

Estimation and Test of Hypothesis's

Problems :

1) Find (a) $t_{0.05}$ when $V = 16$

(b) $t_{-0.01}$ when $V = 10$

(c) $t_{0.995}$ when $V = 7$

Sol:- b) when $V=10$, $t_{-0.01} = -2.764$

a) when $V=16$, $t_{0.05} = 1.746$

c) when $V = 7$, $t_{0.995} = t_{1-0.005} = -t_{0.005} = -3.499$

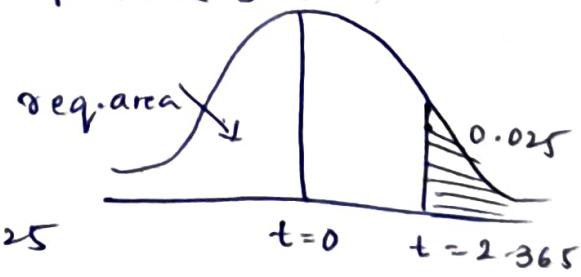
2) Find a) $p(t<2.365)$ when $V=7$

b) $p(t>-1.318)$ when $V=24$

c) $p(-1.356<t<2.179)$ when $V=12$

d) $p(t>-2.567)$ when $V=17$

Sol:- a) when $t<2.365$, $p(t<2.365)$ is given by the area to the left of $t=2.365$



from table

$t_\alpha = 2.365$ for $V=7$ degrees of freedom then $\alpha = 0.025$

$p(t<2.365) = 1-0.025$

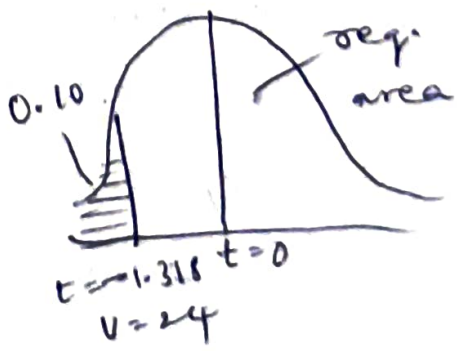
$= 0.975$

b) when $t>-1.318$, $p(t>-1.318)$ is given by the area to the right of $t=-1.318$

From table $t_\alpha = -1.318$ with $V=24$ d.o.f then $\alpha = 0.10$

$\therefore p(t>-1.318) = 1-0.10$

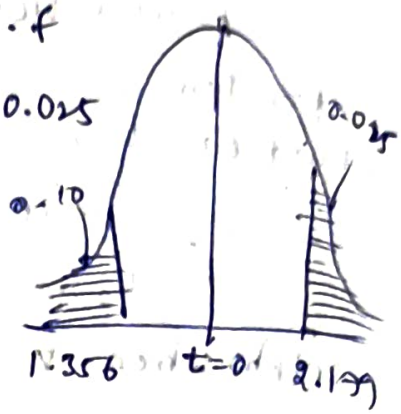
$= 0.9$



c) $p(-1.356 < t < 2.179)$ when $v=12$

when $t < 2.179$ with $v=12$ d.o.f

Area of the right of 2.179 is 0.025



when $t > -1.356$ with $v=12$, dof

Area of the left of -1.356 is 0.10.

when $-1.356 < t < 2.179$, the area is $1 - 0.10 - 0.025$

$= 0.875$

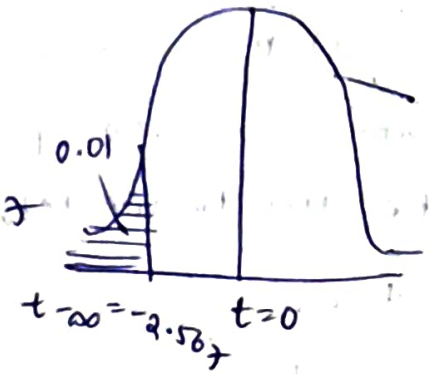
$\therefore p(-1.356 < t < 2.179) = 0.875$

d) when $t > -2.567$

$P(t > -2.567)$ is given by the area to the right of $t = -2.567$

from the table

$t_{-\alpha} = -2.567$ with $v = 17$



3) A sample of size 10 was taken from a population S.D of sample is 0.03. Find the max error 99% confidence.

Solr s = standard deviation = 0.03

n = sample size = 10.

$t_{\alpha/2}$ for $v = 9 = n-1$

$\alpha = 99\%$. then $t_{\alpha/2} = 3.25$

Cor) $\alpha = 1\%$

(i.e $\alpha = 0.01$
$\alpha/2 = 0.005$, $v = 9$)

Max. error

$$t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$= 3.25 \times \frac{0.03}{\sqrt{9}} = 3.25 \times \frac{0.03}{3}$$

$$E = 0.0308$$

(*) Find Unknown population parameter, a statement is made, which is an Estimator. The method of root to determine unknown population parameter is called Estimator.

Ex: Sample mean $\bar{x}$ is Estimator of population because sample mean is a method to determine the population mean.

NOTE 1: A population can have 1 or 2 or many estimatoos.

NOTE 2: The estimator should be selected such that, they are nearer to the parameter value.

NOTE 3: The Estimation can be done in 2 ways.

1) point estimation
2) Interval Estimation.

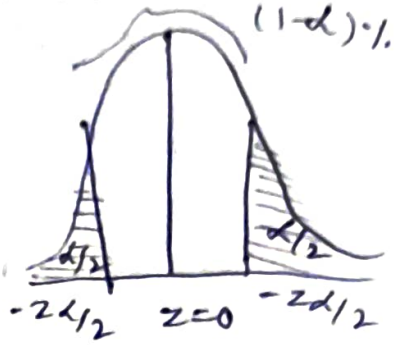
→ An estimate of population parameters is given by a single number is called point Estimation of parameter.

→ An Estimate of population parameter is given by two numbers b/n which the parameter may be consider to lie. is called interval Estimation of parameter.

Eg: If it say that the cost of the bike is 25000.
we are is adding a point estimate

we say that the cost of bike is 25000 to 35000.

we are giving an interval Estimation.



NOTE: Interval estimation of a parameter by the random interval.

i.e confidence interval.

→ The random interval whose end points called cofidence limits

→ whose endpoints are the predetermined such that theeg predetermined numbel is called cofidenu coefficients is statisfied.

→ properties of good Estimator:

And estimate is said to be good Estimator.

① unbaised ② consistence

③ Efficiency ④ Sufficiency.

① Unbaised: The statigatic, $\bar{x}$ is said to be unbaised estimate of θ and μ of

if $E(\theta) = \theta$ (or) $E(\bar{x}) = \mu$.

② An Estimator $\bar{x}_n$ or $\hat{\theta}_n$ of a parameter 'θ' is consistence if it convergence to θ.

③ Efficiency: If $\hat{\theta}_1$ and $\hat{\theta}_2$ to 'a' estimators of a certain parameter θ Then

$\hat{\theta}_1$ is said to be more efficient then $\hat{\theta}_2$ for all sample sizes.

If variance $(\hat{\theta}_1)$ < variance $(\hat{\theta}_2)$

④ Sufficiency: An Estimate the said to be $\bar{x}$ sufficient for a parameter if it contains all the information in the sample

regarding the parameter.

Maximum Error of Estimation :

The sample mean estimate $\bar{x}$ is not always is equal to the mean of the population.

The error, $\bar{x} - \mu$ is the difference b/n Estimator and the quantity which is supposed to estimate

⇒ To find this error, let us make use of the fact that per large 'n'.

$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ is a random variable, having standard distribution.

⇒ It can assigned with probability $1 - \alpha$ that the equality

$$-z_{\alpha/2} < z < z_{\alpha/2}$$

$$-z_{\alpha/2} < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}$$

$$-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{— (A)}$$

$$-(\bar{x} + z_{\alpha/2}\, \sigma/\sqrt{n}) < -\mu < -\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} - z_{\alpha/2} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

(or)

$$\mu \in \left[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}\right]$$

for the population μ.

From (A) $\Rightarrow -z_{\alpha} \frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$|\bar{x} - \mu| < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$M.E = \frac{z_{\alpha}}{2} \frac{\sigma}{\sqrt{n}}$$

Sample size (large samples)

~~The~~ formula for the max. error can also be used.

Determine sample size

$$M.E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$\sqrt{n} = z_{\alpha/2}\frac{\sigma}{M.E}$$

$$n = \left(z_{\alpha/2}\frac{\sigma}{M.E}\right)^2$$

Max error for small samples :

$$M.E = t_{\alpha/2}\frac{s}{\sqrt{n}}$$ where $n-1$ = d.o.f

s = s

n = Sample size

Confidence Interval for μ :

A $(1-\alpha)100\%$ → confidence Interval given by

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

(or)

$$\mu \in \left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

⊙

confidence level $(1-\alpha)$	99%	98%	95%	90%
$z_{\alpha/2}$	2.58	2.33	1.96	1.645

$P(0 < z < z_{\alpha/2}) = 0.45$

$\Rightarrow z_{\alpha/2} = 1.645$

1) A Random sample of size 100 has a SD of 5, what can your say about the max. error with 95% confidence.

Soln W.K.T max. error (M.E) $= z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ —①

Given that $n = 100$ (i.e $n \geq 30$) → large sample.

$s = 5$

confidence level $1-\alpha = 95\%$ (or) $\alpha = 5\%$

$$z_{\alpha/2} = 1.96$$

$$ME = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$= 1.96 \frac{5}{\sqrt{100}} = 0.98$$

2) find the size of sample if the SD of the population is 9. There should be 99% confidence that the error of estimate will not exceed '3'.

Soln Given that $\sigma = 9$

confidence level $1-\alpha = 99\% \Rightarrow \alpha = 1\%$

$$z_{\alpha/2} = 2.58$$

and M.E = 3

$$M.E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$n = \left[z_{\alpha/2} \frac{\sigma}{M.E} \right]^2$$

$$= \left(2.58 \times \frac{9}{3} \right)^2 = 59.9076.$$

$$\boxed{n = 60}$$

Size of sample $n = 60$ //

Test of Hypothesis

Statistical Hypothesis :-

→ We know that the to find unknown population parameter is called Estimation but To decide about population parameter on the basis of simple statisitics. (sample information), If make certain assumptions about the population.

→ These assumptions which true or maynot be true.

→ To Estimate the value of a parameter we should test a Hypothesis about a parameter.

→ The Hypothesis being a reffered as 'H'.

Null Hypothesis :- (H_0)

The Null Hypothesis H_0 is a statistical statement that assumes there is no significant difference or effect b/n the variables ~~mean~~ being study.

Ex:-1) H_0 :- there is no significance reference in student's performance in method A & method B

2) The new medicine as no significant effect compare with Existing treatment.

Alternative Hypothesis :- (H_1)

- The Alternate Hypothesis (H_1) is a statistical Statement that suggests there is a significant reference (or) Effective b/n the variables being the studeied.

Ex:-1) H_0 :- There is no significance reference b/n method A and method B.

H_1 : There is a significance difference b/n method A & B

2) H_0 : The new medicine has no significant effect.

H_1 : The new medicine has a significant effect.

Type 1 Error :

If the thypothesis it is true, But it is rejected in error and is called type 1 Error.

(or)

we can say rejection of the null thypothesis when it is true it is called a type 1 - Error.

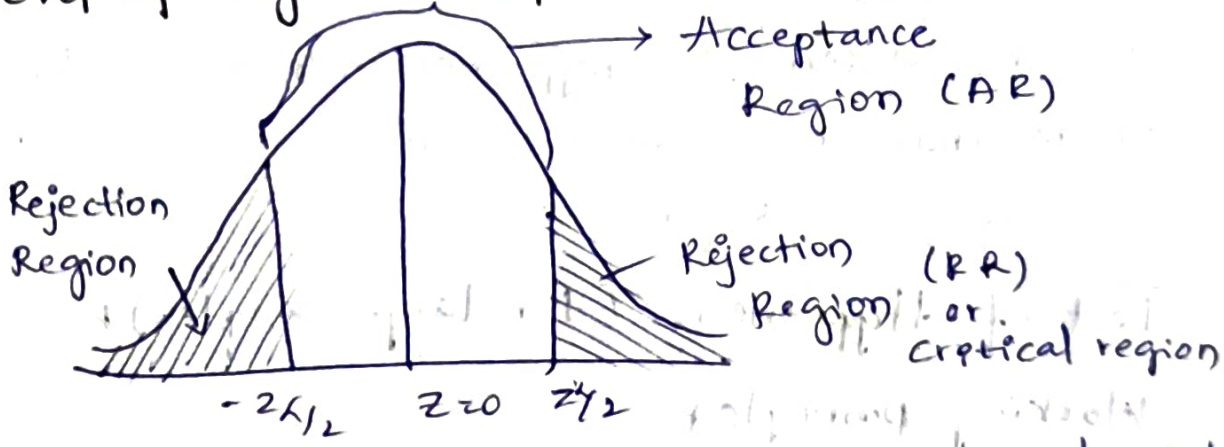
Type 2 - error :

If the thypothesis is false But is accepted in error and it is called Type 2 - Error.

(or)

we can say acceptance of the null thypothesis when it is false it is called Type 2 - Error.

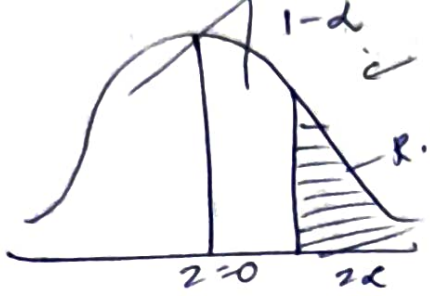
Level of Significance :



⇒ The probability of value of the variate falling in the critical region is the level of significance.

Critical region / Rejection region :

⇒ The critical region is that in which a sample value calling is rejecting



Right-tailed test

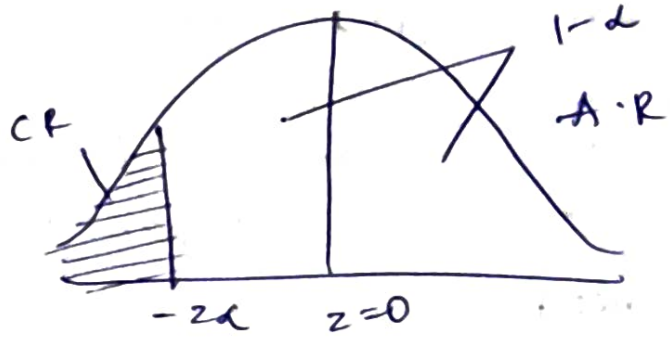
If the Alternative hypothesis H_1 is type of not equal then Critical region lies on both sizes. (Right & left)

⇒ Two-tailed Test (RTT)

Right

If the alternative hypothesis H_1 is the type of greather than the Critical region lies on the right side, as shown in figure.

⇒ Left-tailed Test (LTT)



If the Alternative Hypothesis H_1 is the type of lessthan than the CR ties on left side as shown in figure.

Test of Hypothesis (For large Samples)

Working principle